

Exploration in RL: Contextual Bandit

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CS 1840 Fall 2026**

Announcements

HW2 is out and is due 10/8!

My office hours this week are Thursday, 9:30-11:45, at SEC 6.141.

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Finishing Up From Last Lecture

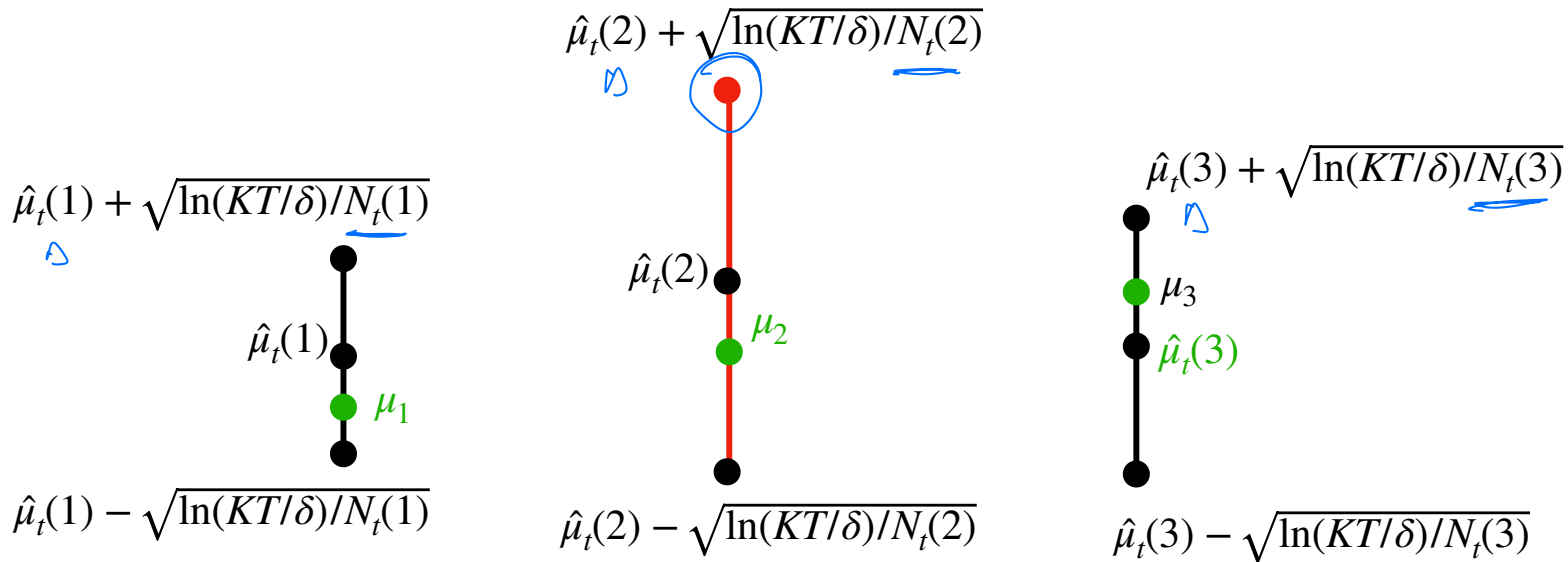


1. The upper Confidence Bound Algorithm

2. Analysis of UCB algorithm

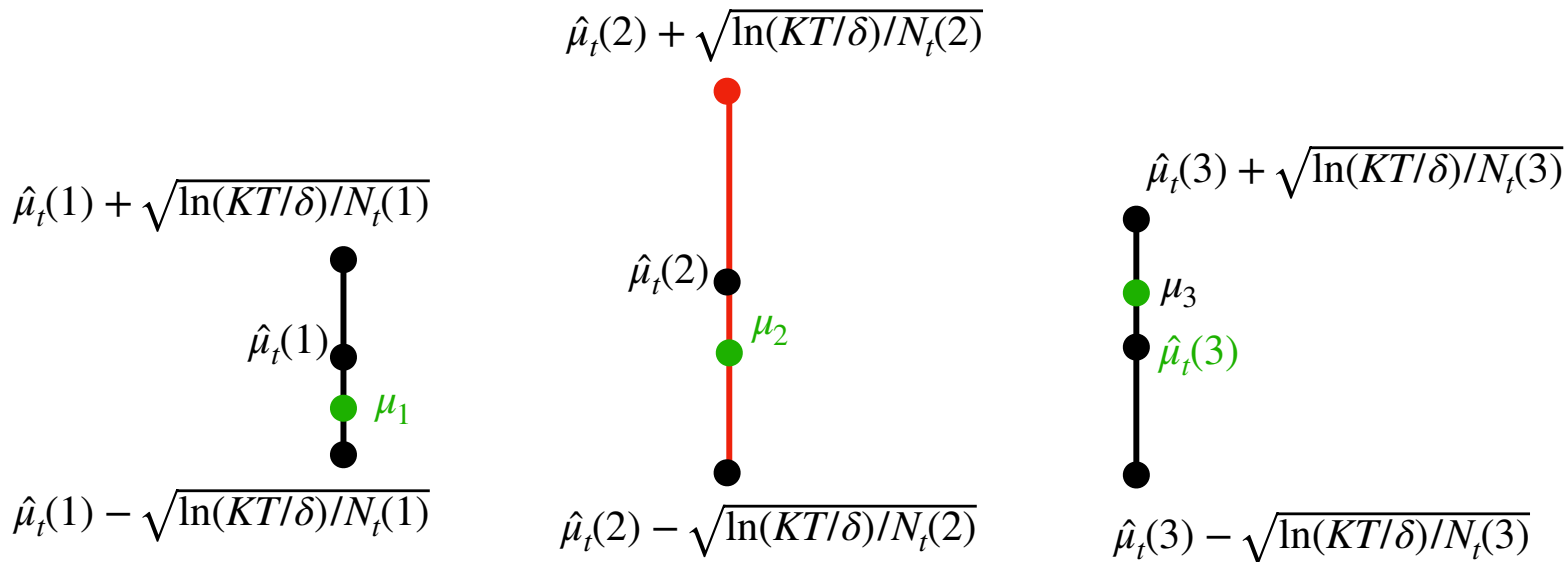
Recap: Intuitive Explanation of UCB

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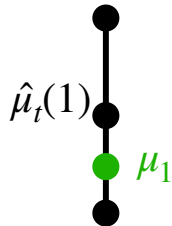
Case 1: it has large conf-interval, which means that it has not been tried many times yet (high uncertainty)



Recap: Intuitive Explanation of UCB

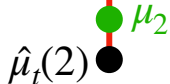
Recap: Intuitive Explanation of UCB

$$\hat{\mu}_t(1) + \sqrt{\ln(KT/\delta)/N_t(1)}$$



$$\hat{\mu}_t(1) - \sqrt{\ln(KT/\delta)/N_t(1)}$$

$$\hat{\mu}_t(2) + \sqrt{\ln(KT/\delta)/N_t(2)}$$



$$\hat{\mu}_t(2) - \sqrt{\ln(KT/\delta)/N_t(2)}$$

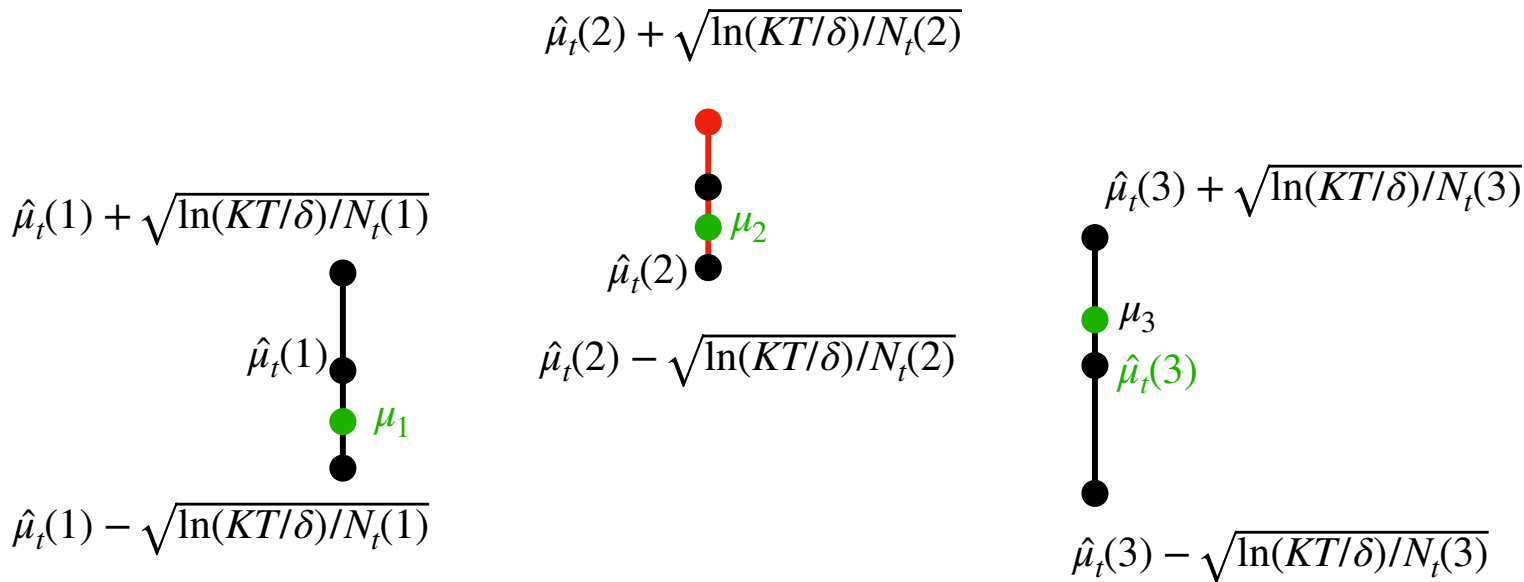
$$\hat{\mu}_t(3) + \sqrt{\ln(KT/\delta)/N_t(3)}$$



$$\hat{\mu}_t(3) - \sqrt{\ln(KT/\delta)/N_t(3)}$$

Recap: Intuitive Explanation of UCB

Case 2: it has low uncertainty, then it is simply a good arm, i.e., its true mean is high!



Recap: Exploration–Exploitation Tradeoff

Case 1: a_t has large conf-interval, which means that it has not been tried many times yet (high uncertainty)

Thus, we do exploration in this case!

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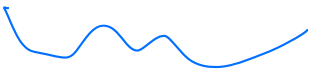
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Case 2: a_t has small conf-interval, then it is simply a good arm, i.e., it's true mean is pretty high!

Thus, we do exploitation in this case!

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

$\uparrow$ mean estimates
 $\uparrow$ half of CI

 arm that has the highest mean

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$$\text{Regret-at-t} = \underbrace{\mu^\star} - \underbrace{\mu_{a_t}}$$

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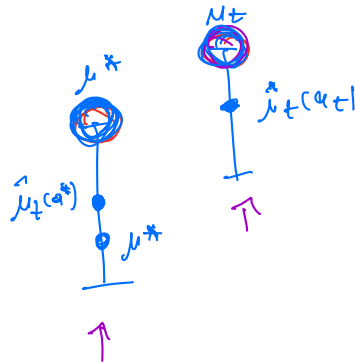
Denote the optimal arm $k^* = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

$$\text{Regret-at-}t = \underline{\mu^*} - \mu_{a_t}$$

Q: why?

$$\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(KT/\delta)}{N_t(a_t)}} - \mu_{a_t}$$

$\underbrace{\hspace{10em}}_{\text{UCB}[a_t]}$



$$\text{UCB}[a_t] \geq \text{UCB}[a^*] \geq \mu^*$$

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$$\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t}$$

$$\mu_{a_t} \geq \hat{\mu}_t(a_t) - \sqrt{\frac{\ln(\dots)}{N_t(a_t)}}$$

LCB

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

$$\begin{aligned} \text{Regret-at-t} &= \mu^\star - \mu_{a_t} \\ &\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t} \\ &\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \end{aligned}$$

length of eI

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$$\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t}$$

$$\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t) \downarrow}} \quad \updownarrow \text{big}$$

Case 1: $N_t(a_t)$ is small

(i.e., uncertainty about a_t is large);

We pay regret, BUT we **explore** here,
as we just tried a_t at iter t !

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

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$$\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t}$$

$$\leq 2 \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \quad \downarrow \text{small}$$

Case 2: $N_t(a_t)$ is large, i.e., conf-interval of a_t is small,

Then we exploit here, as a_t is pretty good (the gap between $\mu^\star$ & μ_{a_t} is small)!

Let's formalize the intuition

Finally, let's add all per-iter regret together:

$$\begin{aligned}\text{Regret}_T &= \sum_{t=0}^{T-1} (\mu^\star - \mu_{a_t}) \\ &\leq \sum_{t=0}^{T-1} 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \quad \sqrt{\frac{a}{b}} = \sqrt{a} \cdot \sqrt{\frac{1}{b}} \\ &= 2\sqrt{\ln(TK/\delta)} \cdot \sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}}\end{aligned}$$

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Lemma (optional):

$$\sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}} \leq O\left(\sqrt{KT}\right)$$


UCB Regret:

[Theorem (informal)] With high probability, UCB has the following regret:

$$\text{Regret}_T = \widetilde{O}\left(\sqrt{KT}\right)$$

(See reading material for more details)

Recap: MAB

Interactive learning process:

For $t = 0 \rightarrow T - 1$

(# based on historical information)

1. Learner pulls arm $a_t \in \{1, \dots, K\}$
2. Learner observes an i.i.d reward $r_t \sim \nu_{a_t}$ of arm a_t

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Arm distributions are fixed across learning..

Question for Today:

How to incorporate contexts into the
interactive learning framework

Outline for today:

1. Introduction of the model

2. Algorithm

3. Theory and some practical considerations

Make the framework Context Dependent:

Interactive learning process:

For $t = 0 \rightarrow T - 1$

$\in \mathbb{R}^d$

1. A new context $x_t \in \mathcal{X}$ appears

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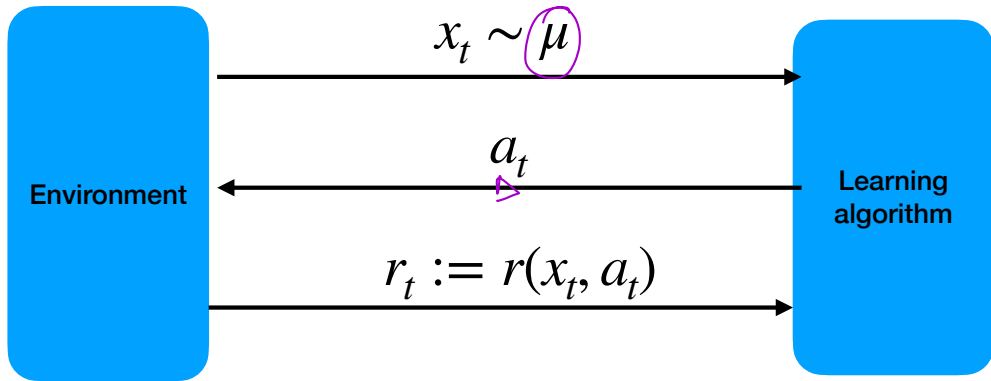
2. Learner picks action $a_t \in \mathcal{A}$

3. Learner observes an reward $r_t := r(x_t, a_t)$

Reward is context and arm
dependent now!

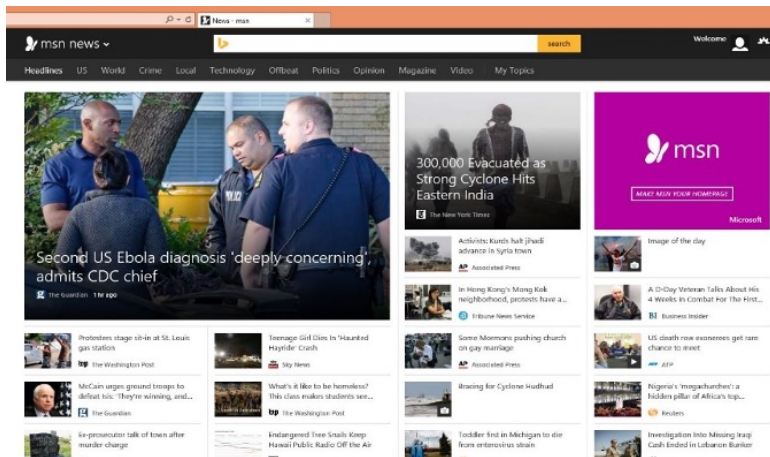
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Examples:

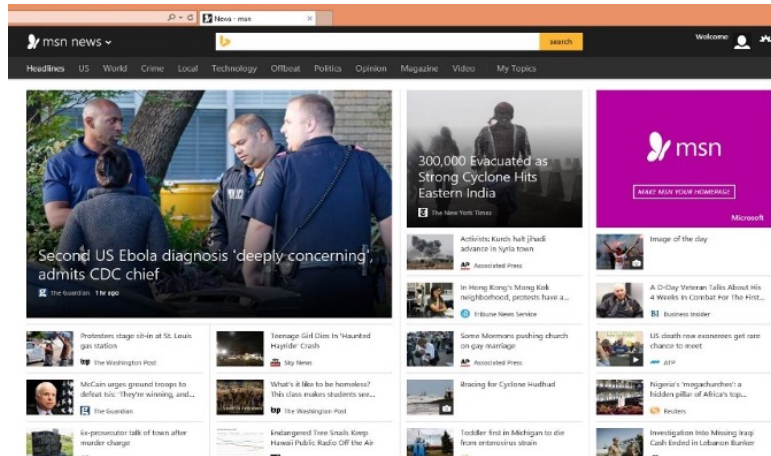
Personalize recommendation system



Examples:

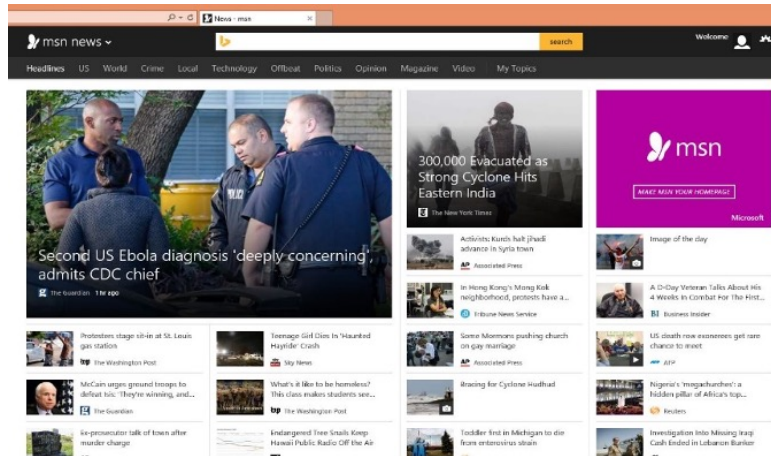
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Context: user's information (e.g.,
history health conditions, age, height,
weight, job type, etc)



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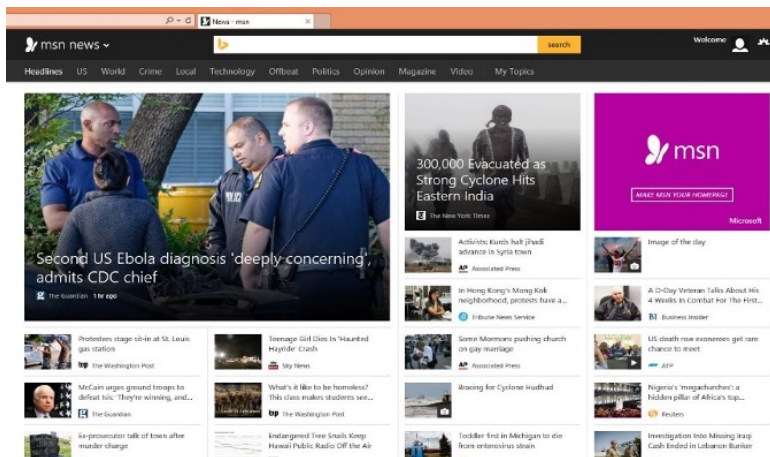


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Decisions (arms): news articles

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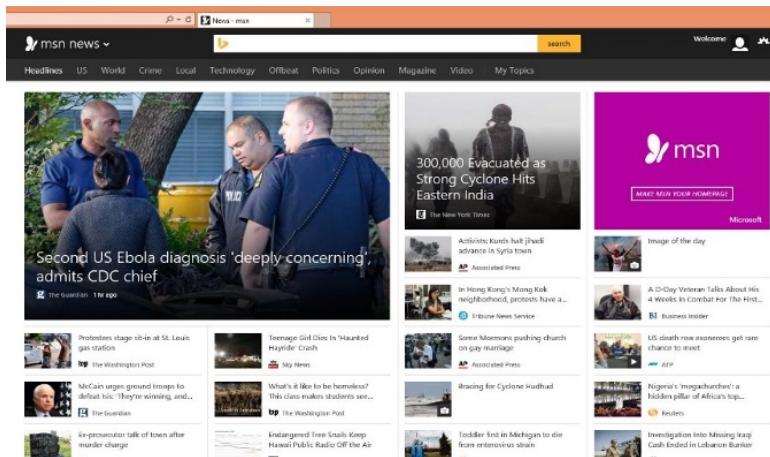
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Goal: learn to maximizes
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Context: user's information (e.g.,
history health conditions, age, height,
weight, job type, etc)

Decisions (arms): news articles

Goal: learn to maximize
user click rate

Different users have different
preferences on news, so
need to personalize

Equivalently, it is an MDP with $H = 1$

Finite horizon MDP with $H = 1$

$$\mathcal{M} = \{\mathcal{X}, \mathcal{A}, r, H = 1, \mu\}$$

$\uparrow$ state space
(context space)

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Finite horizon MDP with $H = 1$

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Objective function:

$$\max_{\pi \in \Pi} \mathbb{E}_{x \sim \mu} [r(x, \pi(x))]$$


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For simplicity, we assume reward is deterministic;
The challenge is really from randomness in contexts

The Regret Metric

Fix a policy class Π (think about π as a classifier from $x \rightarrow a$)

Denote optimal policy $\pi^\star = \arg \max_{\pi \in \Pi} \mathbb{E}_{x \sim \mu} r(x, \pi(x))$

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⏟

Total expected reward if we always
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Total expected reward if we always
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Total expected reward of our learned
sequence of policies

Outline for today:



1. Introduction of the model

2. Algorithm

3. Theory and some practical considerations

Ingredient 1: Importance Weighting

The key challenging here is that we observe $r_t := r(x_t, a_t)$,
but we do not know $r(x_t, a)$ for $a \neq a_t$

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$$\hat{\mathbf{r}} := \begin{bmatrix} 0 \\ 0 \\ \vdots \\ r_t/p(a_t) \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

copying a_t

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$$\boxed{\mathbb{E}_{a_t \sim p} \hat{\mathbf{r}}[a]} = r(x_t, a), \forall a \in \mathcal{A}$$

$$\hat{\mathbf{r}} := \begin{bmatrix} 0 \\ 0 \\ \vdots \\ r_t/p(a_t) \\ 0, \\ \vdots \\ 0 \end{bmatrix}$$

$\checkmark$ $\mathbb{E}[\hat{\mathbf{r}}] = \mathbf{r}(x_t, \cdot)$

Proving Importance Weighting

Assume $a_t \sim p$ ($p \in \Delta(\mathcal{A})$), and we log $p_t(a)$, receive $r_t = r(x_t, a_t)$,

For all $a \in \mathcal{A}$, define $\hat{r}[a] = \frac{r(x_t, a)\mathbf{1}[a = a_t]}{p(a_t)}$, we have: $\mathbb{E}_{a_t \sim p} \hat{r}[a] = r(x_t, a), \forall a \in \mathcal{A}$

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Consider any $a \in \mathcal{A}$:

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$$\begin{aligned}\mathbb{E}_{a_t \sim p} \frac{r(x_t, a)\mathbf{1}[a = a_t]}{p(a_t)} &= \sum_{a_t \in \mathcal{A}} p(a_t) \frac{r(x_t, a)\mathbf{1}[a = a_t]}{p(a_t)} \\ &= \cancel{p(a)} \frac{r(x_t, a)}{\cancel{p(a)}} = r(x_t, a)\end{aligned}$$

$$\mathbb{E}_{a \sim p} [\hat{r}(a)] = r(x_t, a)$$

Ingredient 2: Reward-sensitive Classification Oracle

Recall classic classification:

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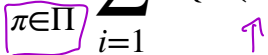
$$\{x_i, y_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, y_i \in \{1, \dots, |\mathcal{A}|\}$$



Ingredient 2: Reward-sensitive Classification Oracle

Recall classic classification:

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Let's generalize it to Reward-Sensitive Classification:

$$\{x_i, \mathbf{r}_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, \mathbf{r}_i \in [0, 1]^{|\mathcal{A}|}$$

↑ ↑

↑

↑

features

vector

Ingredient 2: Reward-sensitive Classification Oracle

Recall classic classification:

$$\{x_i, y_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, y_i \in \{1, \dots, |\mathcal{A}|\}$$

$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ $\leftarrow$ y_i ground truth label

$$\arg \max_{\pi \in \Pi} \sum_{i=1}^N \mathbf{1}\{\pi(x_i) = y_i\}$$

Let's generalize it to Reward-Sensitive Classification:

$$\{x_i, \mathbf{r}_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, \mathbf{r}_i \in [0, 1]^{|\mathcal{A}|}$$

$\begin{pmatrix} x, \\ \Delta \end{pmatrix}$ $\begin{bmatrix} r[1] \\ r[2] \\ \dots \\ r[|\mathcal{A}|] \end{bmatrix}$
Reward label 1
label 2

Ingredient 2: Reward-sensitive Classification Oracle

Recall classic classification:

$$\{x_i, y_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, y_i \in \{1, \dots, |\mathcal{A}|\}$$

$$\arg \max_{\pi \in \Pi} \sum_{i=1}^N \mathbf{1}\{\pi(x_i) = y_i\}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Let's generalize it to Reward-Sensitive Classification:

$$\{x_i, \mathbf{r}_i\}_{i=1}^N, \quad x_i \in \mathcal{X}, \mathbf{r}_i \in [0, 1]^{|\mathcal{A}|}$$

$$\arg \max_{\pi \in \Pi} \sum_{i=1}^N \mathbf{r}_i[\pi(x_i)]$$

$$\begin{bmatrix} r[1] \\ r[2] \\ r[3] \\ \vdots \\ r[|\mathcal{A}|] \end{bmatrix}$$

$$\left(x, \begin{bmatrix} r[1] \\ r[2] \\ \vdots \\ r[|\mathcal{A}|] \end{bmatrix} \right)$$

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We will do reduction to RSC

Summary so far:

1. Importance Weighting: we “magically” get unbiased estimate for all actions!

Assume $a_t \sim p$ ($p \in \Delta(\mathcal{A})$), For all $a \in \mathcal{A}$, define $\hat{\mathbf{r}}[a] = \frac{r(x_t, a)\mathbf{1}[a = a_t]}{p(a_t)}$, we have:

$$\hat{\mathbf{r}} := \begin{bmatrix} 0 \\ 0 \\ \dots \\ r_t/p(a_t) \\ 0, \\ \dots \\ 0 \end{bmatrix}$$

$$\mathbb{E}_{a_t \sim p} \hat{\mathbf{r}}[a] = r(x_t, a), \forall a \in \mathcal{A}$$

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2. Reward-Sensitive Classification:

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Put things together: Explore and Commit

For $t = 0 \rightarrow N - 1$: (# exploration phase)

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For $t = 0 \rightarrow N - 1$: (# exploration phase)

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uniform distribution
over the action space

2. **Uniform-randomly** sample $a_t \sim \text{Unif}(\mathcal{A})$, receive reward $r_t = r(x_t, a_t)$

random action
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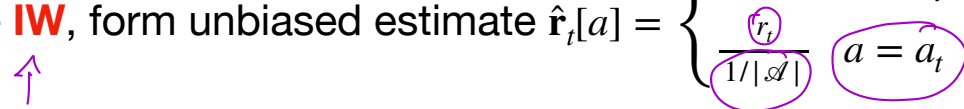
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where $\hat{r}_i \leftarrow \frac{r_i}{1/|\mathcal{A}|}$

$$\hat{\mathbf{r}} := \begin{bmatrix} 0 \\ 0 \\ \dots \\ r_t/p(a_t) \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

$p(a_t) = \frac{1}{|\mathcal{A}|}$

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Call RSC oracle: $\hat{\pi} = \arg \max_{\pi \in \Pi} \sum_{i=0}^{N-1} \hat{\mathbf{r}}_i[\pi(x_i)]$

$$\hat{\mathbf{r}} := \begin{bmatrix} 0 \\ 0 \\ \dots \\ r_t/p(a_t) \\ 0, \\ \dots \\ 0 \end{bmatrix}$$

$\hookrightarrow \pi \in \Pi [r(x, \pi(x))]$

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For $t = N \rightarrow T - 1$: (# exploitation phase)

1. Observe $x_t \sim \mu$, and play $a_t = \hat{\pi}(x_t)$

Outline for today:



1. Introduction of the model



2. Algorithm

3. Theory and some practical considerations

Theory of the Explore and Commit Algorithm

For simplicity, assume Π is discrete (but could be exponential large)

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[Theorem—informal] W/ high probability, properly setting the hyper-parameter N , Explore-and-Commit has the following regret:

$$\text{Regret}_T = T\mathbb{E}_{x \sim \mu}[r(x, \pi^\star(x))] - \sum_{t=0}^{T-1} \mathbb{E}_{x \sim \mu}[r(x, \pi^t(x))] = O\left(\underbrace{T^{2/3} K^{1/3}} \cdot \underbrace{\ln(|\Pi|)^{1/3}}\right)$$

Practical Consideration

Instead of setting a hard threshold for explore and commit, we often interleave explore and exploitation

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ϵ (purple)
 ϵ -greedy:

Every iteration t :

With probability $1 - \epsilon$, we play $a_t = \pi^t(x_t)$,

and w/ probability ϵ , we play $a_t \sim \text{Unif}(\mathcal{A})$

Exploitation

Exploring



Practical Consideration

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$$a \sim p_t, \quad p_t = \underbrace{(1 - \epsilon)\delta(\pi^t(x_t))}_{\Delta} + \epsilon \text{Unif}(\mathcal{A})$$

Put things together: greedy

For $t = 0 \rightarrow \infty$ (# interleave exploration & exploitation)

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For $t = 0 \rightarrow \infty$ (# interleave exploration & exploitation)

1. Observe $x_t \sim \mu$
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4. Via IW, form unbiased estimate $\hat{\mathbf{r}}_t$

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Put things together: greedy

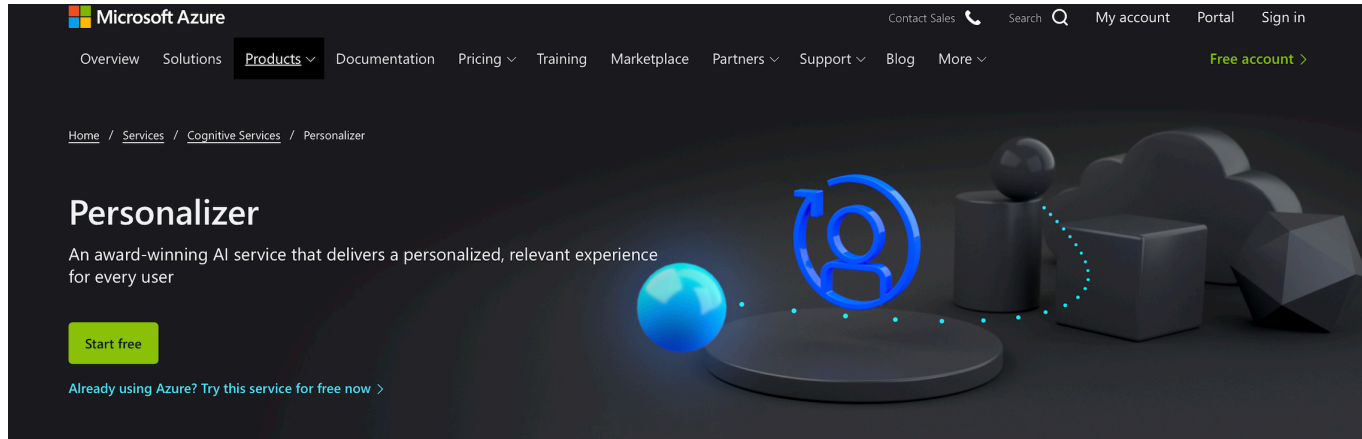
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(Additionally 6. Gradually decay $\epsilon \dots$)

CB algorithm is being used in real world application at Microsoft:

<https://azure.microsoft.com/en-us/services/cognitive-services/personalizer/>



The screenshot shows the Microsoft Azure website's page for the Personalizer service. The top navigation bar includes the Microsoft Azure logo, a 'Contact Sales' button, a search icon, and links for 'My account', 'Portal', and 'Sign in'. Below this is a secondary navigation bar with links for 'Overview', 'Solutions', 'Products' (which is highlighted with a dark background), 'Documentation', 'Pricing', 'Training', 'Marketplace', 'Partners', 'Support', 'Blog', and 'More'. A 'Free account >' link is also present on the right. The breadcrumb trail reads 'Home / Services / Cognitive Services / Personalizer'. The main heading is 'Personalizer', followed by the tagline 'An award-winning AI service that delivers a personalized, relevant experience for every user'. A prominent yellow 'Start free' button is located below the tagline. At the bottom left, there is a link that says 'Already using Azure? Try this service for free now >'. The background of the page features a 3D illustration with a blue '@' symbol containing a person icon, a blue sphere, and various grey geometric shapes like a cylinder, a cube, and a cloud, all set against a dark gradient.

Microsoft Azure

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Overview Solutions **Products** Documentation Pricing Training Marketplace Partners Support Blog More

Free account >

Home / Services / Cognitive Services / Personalizer

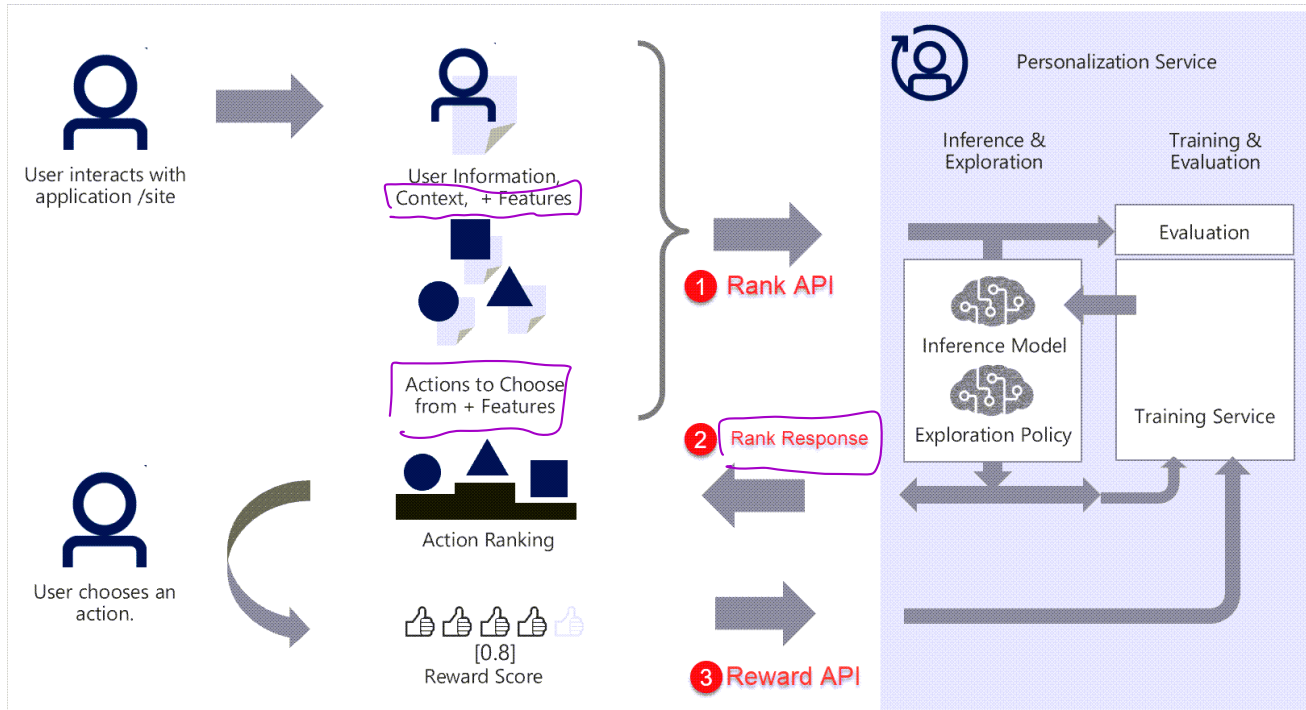
Personalizer

An award-winning AI service that delivers a personalized, relevant experience for every user

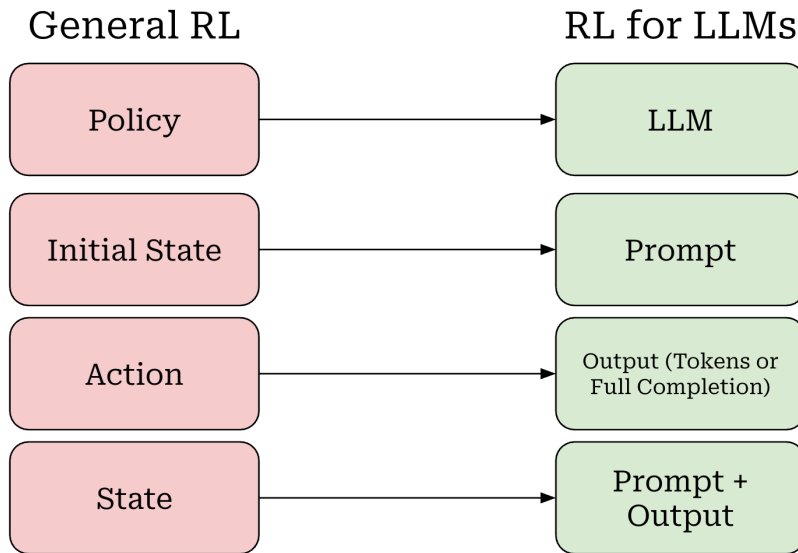
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Framework



CB framework is used to train large language models



Feedback and Attendance

<https://bit.ly/4gYn2nC>

